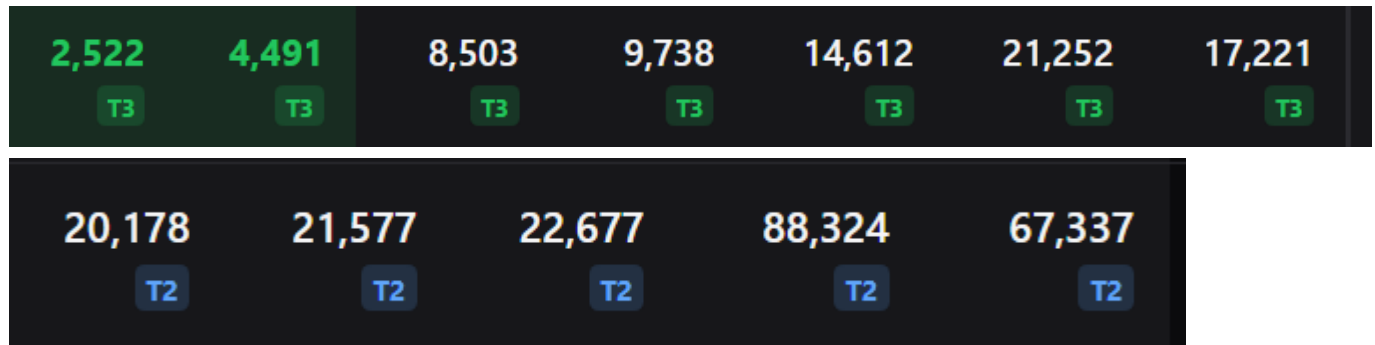


title card

Team name: Avalon Airpot Team

Member: Keming Han, Yecheng Lao, Zixi Wang

slide 1: the picture of our best outcome



slide 2:

solvers and tools: MiniZinc Chuffed and Google OR-Tools CP-SAT

runtime limit = 300s

slide 3: the techniques we used

workflow:

1. use MiniZinc Chuffed to solve out the **tier2** solutions for all data.
2. Warm start to get a solution: we get the tier3 solution based on the solution of tier2.
 1. fix all the **gate** got from the existing tier2 solution
 2. recreate vars of **task_start** and **task_labor**
 3. use tier2 value as CP-SAT hint:
 1. try the solution got before
 2. if failed, release 10% of the var **task_start** and **task_labor**, and resolve. The released rate can be self-defined by the user. We have tried 10% to 30%.
3. optimisation(LNS): as if we get a solution, we can fix 90% of the solution and on this base, searching for a better solution with the added **constraint: cost < current_cost;**
 1. **question:** if the released percentage is too low, then we cannot find a new solution; if too much, then the runtime may still exceed the runtime limit.